

1

Simple Molecules: Hybridization, Conformation and Configuration

Aims

By the end of this chapter you should be familiar with:

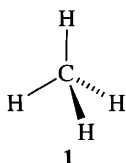
- The representation of three-dimensional molecules in two dimensions
- The hybridization of carbon in alkanes, alkenes and alkynes
- The energy profile associated with conformational changes in ethane as the carbon–carbon bond rotates about its axis
- Eclipsed, staggered, *gauche*, *cis* and *trans* conformations as applied to, for example, propane and butane
- Chair and boat conformations, and ring inversion of cyclohexane
- 1,3-Diaxial interactions in substituted cyclohexanes

1.1 Introduction

Chemistry, like everyday life, takes place overwhelmingly in three dimensions. Stereochemistry embraces the spatial aspects of chemistry and can be considered in two parts. The first deals with the shapes and properties of mainly three-dimensional molecules and involves a knowledge of the terms **conformation**, **configuration** and **chirality**, which are introduced in the first two chapters. The second aspect deals with reactivity and includes the preferred or obligatory direction of approach of reagents, and also the consequences for the nature of the products. In respect of reactivity, it can be said that except for spherical reactants, *e.g.* H^+ and Cl^- , there is almost always a preferred direction of approach of one molecule or ion toward another.

We consider first the simplest organic molecule methane, CH_4 , in

The representation of methane in two dimensions is given.



particular its geometry, its representation in two dimensions and its make-up from the component atoms. In methane, all four carbon–hydrogen bonds are equivalent and are directed toward the corners of a regular tetrahedron with carbon at the centre. The HCH bond angles are all $109^{\circ}28'$, a value that is found only in molecules with ‘tetrahedral’ symmetry. The normal depiction of methane is given in **1** and the convention, which is applied generally, for representing this three-dimensional molecule in two dimensions is now described. The central carbon is taken to lie in the plane of the paper, together with any two hydrogens. Most commonly these are the upper and left hydrogens and, as shown in **1**, bonds between each of these two hydrogens and carbon are represented by lines of ‘normal’ thickness. The thick ‘wedge’ represents a bond between carbon and the hydrogen in front of the page, and the dashed line indicates a bond between carbon and the hydrogen behind the page.

Note that any three atoms must lie in a plane. One can always draw a plane through three points; in the everyday world, for example, a three-legged stool never rocks. The three atoms that lie in the plane of the paper in **1** are chosen for convenience.

1.2 Hybridization: Methane

When a bond is formed between, say, C and H, the electronic distribution in the orbitals is perturbed and two atomic orbitals that normally each contain an electron are said to overlap. In the case of a simple molecule such as in methane, the question arises: how does one reconcile the equivalence of the four C–H bonds in methane with the outer shell electron configuration, $2s^2 2p^2$, of carbon? Hydrogen contributes its one electron in a $1s$ orbital, whereas carbon has four atomic orbitals of appropriate energy available for bonding. These four orbitals are one $2s$ and three $2p$ orbitals, which are designated $2p_x$, $2p_y$ and $2p_z$ according to their different directions in space; x , y and z refer to the directions of a Cartesian coordinate system. A $2s$ and a $2p_y$ orbital are shown in Figure 1.1.

The $2s$ orbital is spherically symmetrical, and its electron density is highest at the nucleus. Each $2p$ orbital is cylindrically symmetrical

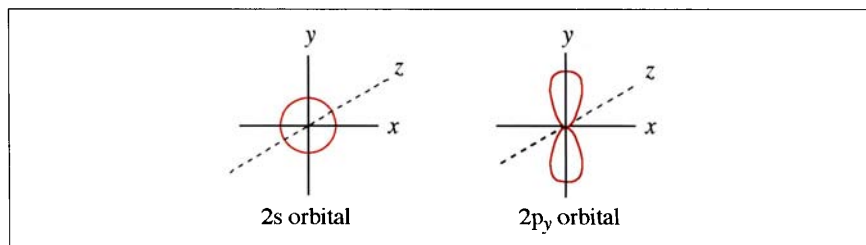


Figure 1.1

and, in contrast to its 2s counterpart, has zero electron density at the nucleus. If one 2s and three 2p orbitals were used without modification in bonding, then four equivalent bonds would not be found in methane.

A slightly higher energy state of carbon is described by the electron configuration $1s^2 2s^2 2p^3$, and from the 2s and the three 2p orbitals four equivalent mixed or 'hybrid' orbitals are obtained. In the case of carbon in methane, and saturated carbon in most other molecules, the **hybridization** is described as sp^3 . These orbitals have some characteristics of the component atomic orbitals. Accordingly, an sp^3 orbital has finite charge density at the nucleus, as does the 2s (but not the 2p) orbital; also it has directionality, in common with the 2p orbitals, with lobes now of unequal size.

The structure of methane that indicates the equivalent sp^3 hybrid orbitals of carbon bonding with four hydrogens is shown in Figure 1.2. The observed HCH bond angles in methane are such as to place the hydrogen atoms as far apart as possible. The same angles are observed in, say, CF_4 , but in general the C–C–C bond angles in acyclic saturated molecules are slightly greater, and an example is propane where the value is *ca.* 112° .

Four equivalent C–H bonds are formed from sp^3 hybrid orbitals, themselves derived from one 2s and three 2p orbitals.

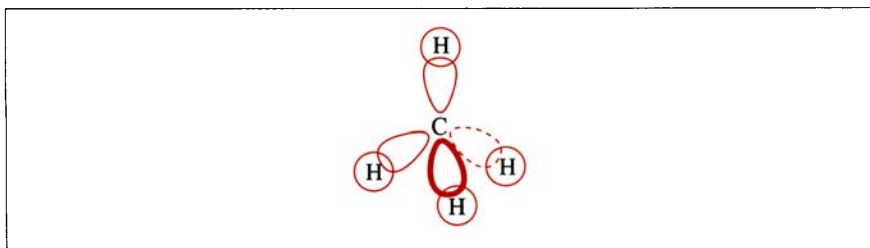


Figure 1.2

1.3 Hybridization: Ethene and Alkenes

Analysis of the geometry of methane does not explain that of ethene, C_2H_4 , which contains two equivalent carbons. From electron diffraction data (see Thompson¹), ethene is known to be a flat molecule with all six atoms in a plane and with bond angles of 120° . The bonding of ethene can be rationalized by using the same orbitals, one 2s and three 2p ($2p_x$, $2p_y$, $2p_z$), but with the difference that *one* of the 2p orbitals does not participate in the hybridization. The net result is that there are three sp^2 hybrid orbitals and one p orbital per carbon. There are a number of significant differences about bonding in ethene compared to that in methane. The sp^2 hybrid orbitals in ethene and indeed all alkenes possess a greater percentage 's' character, and consequently the electrons reside closer to the carbon nucleus. Additionally, the change of hybridization brings about a change of bond angle. In ethene, and around the sp^2

Three sp^2 bonding orbitals are formed from one 2s and two 2p orbitals. The remaining p orbital on each carbon of ethene is used to form the π bond.

hybrid carbons of alkenes, the bond angles are 120° , which leads to the flat structure. This geometrical arrangement means that the substituents around the alkene carbons are as far away from each other as possible, a feature that is true also of methane (see Section 1.2). It follows that in cyclohexene, for instance, four of the six carbon atoms (and two hydrogens) are coplanar.

What then of the p orbitals that did not participate in the formation of sp^2 hybrid orbitals? These remain as p orbitals, one at each carbon, and each p orbital contains one electron. Together they form a bond, termed a π bond, between the carbon atoms. The mode of overlap is 'sideways on'. This contrasts with the end-on overlap that results in formation of σ bonds, and which in the present context involves C–H bonds in methane (Figure 1.2) and ethene, and the other bond between carbon and carbon in ethene.

The sp^2 hybrid orbitals in ethene and the p orbital are shown in Figure 1.3. The sp^2 hybrid orbitals form one carbon–carbon σ bond and two carbon–hydrogen σ bonds. Ethene may be drawn as in Figure 1.4a; in Figure 1.4b ethene is drawn to highlight the overlap of the p orbitals of the two carbons that give rise to the π bond.

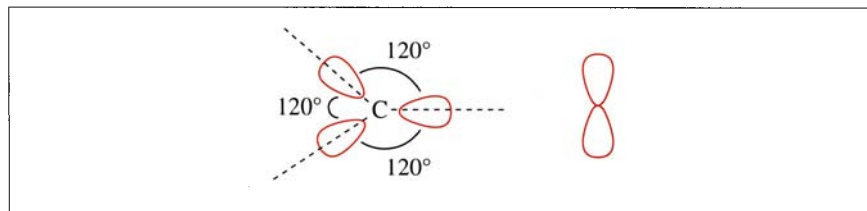


Figure 1.3

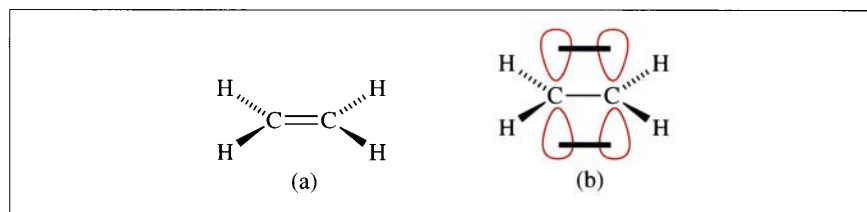


Figure 1.4

The presence of the π bond confers properties on an alkene that mark it out as different from an alkane. In particular, the π bond, by the nature of its sideways overlap of the constituent p orbitals, is weaker than a σ bond. Moreover, the electrons of the π bond are relatively exposed, above and below the plane of the alkene. These electrons are the source of reactivity of the alkene toward electrophiles, as in, say, electrophilic addition of bromine (Chapter 4). The π bond in ethene (and other alkenes) is, however, sufficiently strong that it prevents rotation around the carbon–carbon σ bond, which is a well-documented property of the carbon–carbon bond in ethane (Section 1.6). The bonding between sp^2

The π bond of alkenes reacts with electrophiles, e.g. bromine.

hybrid carbons of an alkene consists of one σ bond and one π bond; collectively, these are referred to as a double bond.

Double bonds are also formed between carbon and nitrogen to give imines ($R_2C=NR$), and also between carbon and oxygen to give aldehydes and ketones. In both these cases also there is one σ bond and one π bond. In aldehydes, ketones and imines, oxygen especially, and nitrogen, are more electronegative than carbon. Accordingly, the electrons of the π bond in particular are drawn toward the oxygen and the nitrogen, respectively. This has the effect of making oxygen and nitrogen partially negatively charged, and the carbons of these double bonds are correspondingly partially positive.

1.4 Hybridization: Ethyne

For completeness, we consider briefly the triple bond in ethyne (acetylene, $HC\equiv CH$), though its stereochemical significance is limited. The carbons of ethyne utilize one s and one p orbital to form two sp hybrid orbitals. This leaves two unhybrid p orbitals. The sp hybrid orbitals are used to form the σ bonds, one to the other carbon, and one to hydrogen. The p orbitals at each carbon form two separate, and independent, π bonds that are at right angles to each other as viewed along the axis of the molecule. Such π bonds are called **orthogonal**. Bonding in nitriles, $RC\equiv N$, is similar with the triple bond again made up of one σ bond and two orthogonal π bonds. The bonding in ethyne is shown diagrammatically in Figure 1.5.

Ethyne sp hybrid orbitals form two σ bonds; two remaining p orbitals on each carbon form orthogonal π bonds.

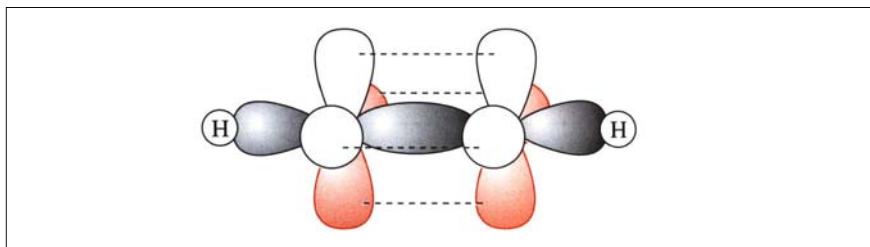


Figure 1.5

Along the series ethane (carbons sp^3), ethene (sp^2) and ethyne (sp) there is an increase in the percentage s character of the carbon hybrid orbitals. This implies a greater electron density at the carbon nucleus as one moves along this series, which is in accord with the progressively shorter carbon-carbon bond lengths observed. Average values of 0.154 nm (ethane), 0.133 nm (ethene) and 0.12 nm (ethyne) have been found. Table 1.1 lists some bond angles in hydrocarbons.

Table 1.1 Bond angles in hydrocarbons^a

Bond type	Bond angle
C–C–C	109°28' (ideal) 111–112° (found)
C=C–H	120°
C≡C–H	180°

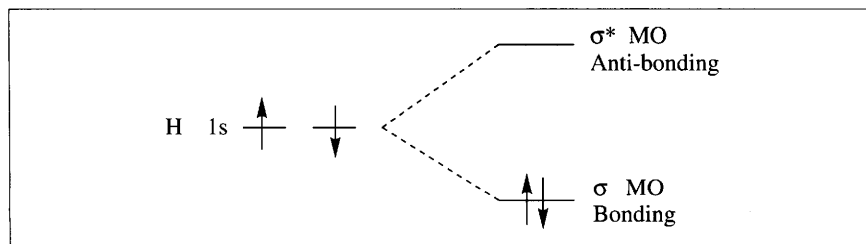
^aValues can vary by a few degrees from those cited

1.5 Bonding and Anti-bonding Orbitals

A further general point about molecular orbitals is illustrated with the hydrogen molecule. This is composed of two hydrogen atoms, each of which contributes one electron from a 1s orbital to form the molecular orbital. Two electrons in this molecular orbital form the σ bond in the hydrogen molecule. It may appear that two atomic orbitals, that together are *capable* of containing four electrons, form one molecular orbital that is only capable of containing two electrons. The combination of atomic orbitals is slightly more involved.

When two atomic orbitals combine they produce two **molecular orbitals** of unequal energy (Scheme 1.1). The lower energy molecular orbital is lower in energy than the component 1s atomic orbitals; it is called a **bonding molecular orbital** (MO) and contains the two electrons that form the σ bond. Additionally, a second molecular orbital is formed that is higher in energy than the 1s orbitals, and in the ground state hydrogen molecule it is unoccupied. This higher energy orbital is known as an **anti-bonding orbital** and in this case is given the symbol σ^* .

When two atomic orbitals produce two molecular orbitals, one is lower in energy (bonding), and the other is higher in energy (anti-bonding) than the atomic orbitals.

**Scheme 1.1**

Similar behaviour is also shown by p orbitals, which form π and π^* molecular orbitals.

1.6 Conformation: Ethane

Unlike methane with its fixed geometry, ethane is not rigid and rotation occurs around the carbon–carbon bond. Conformations of a molecule are

2

Chiral Molecules: One Stereogenic Centre

Aims

By the end of this chapter you should be familiar with:

- All the terms in coloured bold type
- The relationship between enantiomers
- The principles of polarimetry
- Fischer projections
- The CIP *R/S* configuration nomenclature
- Racemization

2.1 Chirality, Enantiomers and Optical Activity

When an sp^3 hybridized tetrahedral carbon is attached to four different groups, as in **1** (see below), the molecule cannot be superimposed on its mirror image. This is the same property that a right hand, say, placed in front of a plane mirror possesses. Molecules such as **1** are said to be **chiral** (or handed), from the Greek word 'cheir' for hand.

Chirality then is the property of handedness, and the adjective 'chiral' refers to the molecule as a whole, rather than to a particular atom. Molecules such as CCl_4 and CH_2Cl_2 that can be superimposed on their mirror images are said to be 'achiral'.

Molecules such as **1** and **2** are termed 'enantiomers', from the Greek 'enantio' meaning opposite. **Enantiomers** are defined as a pair of molecules related as *non-superimposable* mirror images; note that it is essential to include 'non-superimposable' in the definition. It can be seen that enantiomers are a particular sub-class of stereoisomers. Each enantiomer of a pair has the same physical properties (*e.g.* m.p., b.p. and solubility) with one exception. This distinguishing feature arises when, in separate experiments, plane polarized light is passed through a solution of each

enantiomer in the same solvent, and using the same cell. If the concentration of the two enantiomers is the same, then the plane of polarized light is rotated in opposite senses and by the same amount. This difference in behaviour has important diagnostic value.

Although this chapter deals with chirality in certain molecules, one should be aware that chirality is to be found in the wider world. As mentioned above, the word chirality comes from the Greek for hand. A hand, and a foot, are chiral and fulfil the same criteria as chiral molecules. Accordingly, a left hand placed in front of a mirror gives a right hand as its mirror image (Figure 2.1). These cannot be superimposed, as can readily be demonstrated with one's own hands. Also, a glove is chiral and a pair of gloves consists of two enantiomeric gloves. Likewise, a shoe or a sandal is chiral. A plain sock is not chiral but it will of course readily adapt to the chirality of the foot.

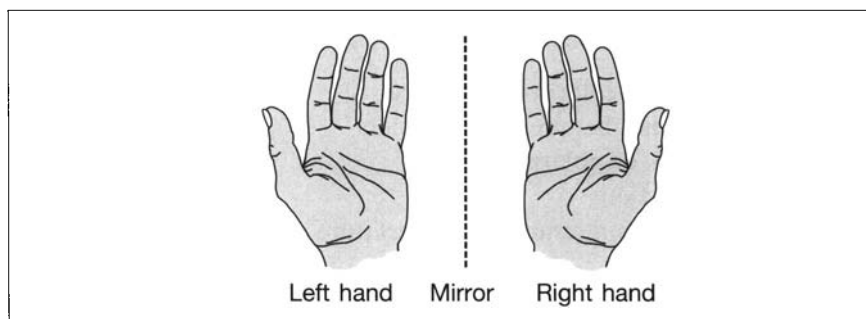


Figure 2.1

Another common chiral object is a cylindrical helix, such as a bed-spring. A helix with its mirror image is shown in Figure 2.2 and again object and mirror image are not superimposable. The thread of a screw ensures that it too is chiral. The body shell of a car as it enters the assembly line is not chiral, and it has a plane of symmetry. When the car leaves the assembly line it has acquired, for example, a steering wheel on one side or another. The finished car is now not superimposable on

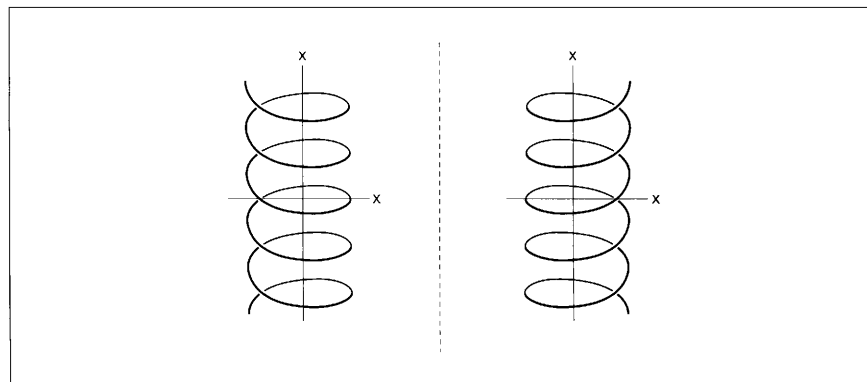


Figure 2.2

its mirror image, and it is therefore chiral; it has also lost its plane of symmetry.

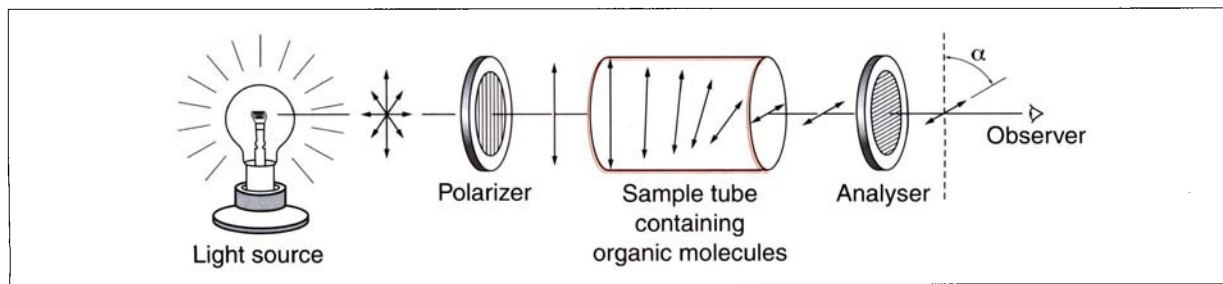
A rowing boat is clearly not chiral, but note that it possesses a plane of symmetry. An engine, presumed symmetrical, fitted to this boat will have a propeller, almost certainly with three blades. The propeller itself is chiral (verify this by observation if possible) and after the engine with its propeller is fitted, the boat as a whole has now become chiral by virtue of its propeller, which has also caused the boat to lose its plane of symmetry.

Some types of aircraft use one or more propellers that each have four blades; each propeller is again chiral. Is a jet engine chiral? (Consider the engine, not running, viewed from the front of the plane).

Light may be regarded as a wave motion that contains oscillating electric and magnetic fields. The electric field of light oscillates in all planes at right angles to the direction of propagation of the light wave. When a beam of normal light is passed through a device called a **polarizer**, that in effect acts as a filter, only light waves that are oscillating in a single plane are allowed to pass through. The polarizer thus serves to block the passage of light that is oscillating in all other planes. The light that emerges from the polarizer is said to be plane polarized.

In 1815 the French physicist Biot made the important observation that when a beam of **plane polarized light** is passed through a solution of certain naturally occurring organic compounds, the plane of polarized light is rotated either to the left or to the right. Molecules that induce this rotation are said to be **optically active**.

The simplest class of optically active molecules has one carbon bonded to four different substituents, and of course it is necessary for one enantiomer to be in excess over the other. Measurements of optical activity are carried out with a polarimeter, which is shown schematically in Figure 2.3. The direction and extent of the rotation of the plane of polarized light is the basis of specific rotation, which is dealt with in this section.



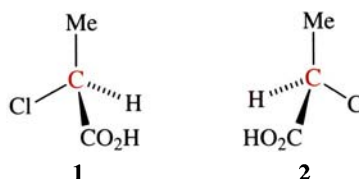
Each enantiomer undergoes chemical reactions at the same rate as long as the other reagent (and the solvent) is not chiral. However, in the

Figure 2.3

Enantiomers are chiral molecules that are non-superimposable mirror images of each other, and with one exception have identical physical properties. Enantiomers react at the same rate with achiral reagents.

case in which the other reagent is chiral, the two enantiomers will react at different rates; this is the basis for enzymic selectivity, which is discussed later (Section 3.8).

A pair of enantiomers is represented by **1** and **2**, in which the carbons shown in colour are called '**stereogenic centres**'. They are also less justifiably called 'chiral centres'; the term 'asymmetric carbon' is now encountered less frequently.



The opposite configurations of the stereogenic centres in **1** and **2** are indicative of the opposite directions in which they rotate the plane of polarized light. However, it is not possible from inspection of a particular enantiomer to say in which direction the plane is rotated. Enantiomers that rotate the plane to the right, *i.e.* clockwise, are called **dextrorotatory** and are sometimes given the symbol '*d*'; those that rotate to the left, anti-clockwise, are correspondingly called **laevorotatory** with the symbol '*l*'. It should be made clear that the terms dextrorotatory and laevorotatory are defined with the observer looking into the propagating beam. These names and their symbols are empirical and have no fundamental significance. Also, those compounds that rotate the plane of polarized light to the right are frequently given the symbol (+), and those that rotate to the left the symbol (−).

In order to characterize the extent of rotation, an index called **specific rotation**, $[\alpha]_D$, is used; this is given by equation (1):

$$[\alpha]_D^t = \frac{100\alpha}{l \times c} \quad (1)$$

In equation (1), ' α ' (alpha) refers to the observed rotation in degrees, '*l*' to the length of the polarimeter cell in decimetres, and '*c*' refers to the concentration of the sample in g 100 cm^{−3}, *i.e.* g per 100 cm³. On the left-hand side of the equation, '*t*' refers to the temperature in °C at which the measurement takes place, and the subscript 'D' indicates that light from the D line of a sodium lamp with a wavelength of 589.6 nm was used. Sometimes the concentration *c* is expressed as g cm^{−3}; in this case, equation (2) is employed:

$$[\alpha]_D^t = \frac{\alpha}{l \times c} \quad (2)$$

If a measurement is made as a check on a published value of the spe-

cific rotation, care must be taken to ascertain whether the concentration relates to equation (1) or equation (2). For example, a reasonable concentration, such as 2 g in 100 cm³ in respect of equation (1), might well exceed the solubility of the compound if taken to be concentration $c = 2$ g in 1 cm³, as used in equation (2).

Values of specific rotation are expressed as in equation (3) for the amino acid cysteine (compound **30** in the Problems section of this chapter):

$$[\alpha]_{\text{D}}^{25} = -16.5 \text{ (conc. 2 g in 100 cm}^3 \text{ H}_2\text{O)} \quad (3)$$

cysteine

The units of $[\alpha]$ are 10⁻¹ deg cm² g⁻¹ but are not usually included; until recently, values were expressed in deg. The conditions used are given because $[\alpha]$ can vary with (a) wavelength (sometimes the incident light is from a mercury lamp and has wavelength 546.1 nm.), (b) temperature and (c) solvent; effects of polarity of the solvent and hydrogen bonding between solvent and substrate can affect $[\alpha]$ markedly.

The use of g cm⁻³ to express concentration in equation (1), though common in polarimetry, is unusual; concentration is more commonly expressed in mol dm⁻³. A molar rotation $[M]$ is known, but is used less frequently.

The example shown by **1**, *i.e.* a stereogenic carbon bonded to four different groups, is the most common reason for a molecule to show chirality, and of course there can be several stereogenic centres in a molecule. However, the need for a stereogenic centre is not an absolute requirement for a molecule to be chiral, and this is considered in Chapter 5. All cases must, however, fulfil the fundamental condition: for a molecule to be chiral and to display optical activity, it must be non-superimposable on its mirror image.

The stereochemistry that is possible at saturated carbon is also discussed by McMurry.¹

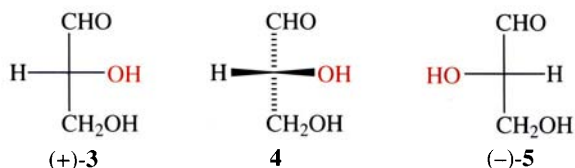
Enantiomers rotate the plane of polarized light in opposite directions by an amount that is characteristic of the compound. Specific rotation, $[\alpha]$, is a measure of the rotation at unit concentration using a cell (that contains the solution) of unit length.

2.2 How to Specify a Configuration

2.2.1 Fischer Projections

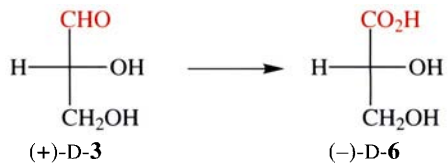
Two conventions are in use to describe the configuration of a stereogenic centre. The older system uses the symbols D and L, and is still employed for two important classes of compound, namely amino acids and sugars; otherwise it has been superseded. The D/L system will be considered first. The use of capital letters here serves to distinguish this convention from *d/l* (page 22), which relates only to the sense in which the plane of polarized light is rotated.

In the early 1900s it was not possible to assign an absolute configuration to a chiral molecule with even a single stereogenic centre, although the need for a system of assignment was appreciated. Accordingly, an arbitrary standard was put in place; this consisted of the molecule (+)-glyceraldehyde (**3**), selected because of its close relationship to sugars. The representation of the three-dimensional structure was addressed as follows.



(+)-Glyceraldehyde was drawn as **3**, using what is known as a **Fischer projection**. The convention is that a stereo drawing of this molecule is shown as in **4**. The vertical bonds are directed behind the page from the stereogenic carbon (which is not shown in a Fischer projection) and the horizontal bonds in **3** resemble a bow tie in **4** projecting out of the page. Fischer was the second Nobel laureate in chemistry in 1902.

The decision was taken to assign (+)-glyceraldehyde (**3**) the absolute configuration shown, which was termed 'D', and the enantiomer (–)-glyceraldehyde was then represented by **5**, and given the symbol 'L'. The terms D and L refer to the molecule as a whole. In 1951, Bijvoet and his group in Utrecht showed by X-ray structure analysis that the original arbitrary assignments, happily, were correct. This finding was significant because the configurations of many compounds had been designated by relating them to (+)-glyceraldehyde. One example involves the configuration of (–)-glyceric acid (**6**), obtained by oxidation of (+)-glyceraldehyde [typically with mercury(II) oxide]. Since the oxidation has taken place away from the stereogenic centre and the oxidizing agent is mild, one can conclude that there has been no change in configuration of the stereogenic centre during formation of **6** from **3**.

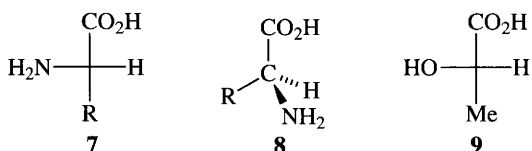


The configuration of (–)-**6** is as shown and is accordingly termed D-glyceric acid. This result shows that there is no correlation between configuration and the sign of rotation.

Except for glycine, the alpha amino acids derived from proteins of higher living organisms have at least one stereogenic centre. Those with a single such centre are represented by Fischer projection formulae **7** (or the corresponding stereo diagram **8**), and all have the L configuration shown at the alpha carbon. In these amino acids, *e.g.* serine (**7**, R = CH₂OH), assignment of configuration was achieved by chemical conversions that related it to L-(–)-glyceraldehyde using L-lactic acid (**9**) as a relay.

Amino acids of type **7** of course may be assigned configurations in the *R/S* convention (Section 2.2.2.); after having read this section, attempt question **2.6** in the Problem section (page 35) to ascertain whether the four amino acids **28–31**, which are all of L configuration, are all described by *R* or *S*.

Fischer projections, used for naturally occurring amino acids and sugars, are associated with symbols D and L. The former indicates the same configuration as the standard, D(+)-glyceraldehyde (**3**).



There are further rules for the use of Fischer projections: (a) where relevant the vertical direction is that of the carbon chain; (b) in the vast majority of cases the carbon in the highest oxidation state is located at the top of the diagram.

Care must be exercised in testing whether two structures are, or are not, superimposable. Thus, a Fischer projection diagram of a chiral molecule with one stereogenic centre may be rotated only through 180° around the central carbon with preservation of configuration. Rotation of the Fischer projection by 90° inverts the stereogenic centre to give the other enantiomer; exchange of any two groups also produces the other enantiomer. A discussion of the D/L convention is given by Buxton and Roberts.²

Verify these rules with a molecular model of **3**.
Verify that if one takes a Fischer projection of, *e.g.* **3**, and exchanges any two groups, one produces the enantiomer.

A limitation of the D/L system may be seen from the example of (+)-camphor (**10**), which was assigned configuration D. Comparison with the formula **3** for D-glyceraldehyde reveals that the inter-relationship is, at best, tenuous. Some suppliers still offer D-camphor for sale.



2.2.2 The Cahn–Ingold–Prelog *R/S* convention

Cahn, Ingold and Prelog developed a new convention for assignment of configuration at a stereogenic centre (and other more involved cases). Cahn was editor of the then *Journal of the Chemical Society*, Ingold from University College London initiated and established the study of organic reaction mechanisms, and Prelog from ETH Zurich was a distinguished organic chemist and Nobel laureate in 1975. Frequently referred to by the initials of its proponents, the **CIP convention** (also known as the *R/S* convention) was proposed in 1966,³ following an earlier paper by Cahn and Ingold in 1951.⁴ The convention is now in almost universal use and forms part of the IUPAC (International Union of Pure and Applied Chemistry) rules of nomenclature.⁵ In a clean break with previous symbolism the symbols '*R*' (Latin, 'rectus' = right) and '*S*' (Latin, 'sinister' = left) were adopted. Interestingly, the symbol '*R*' was taken from the Latin word for 'right' in the sense of 'just' or 'correct', since the Latin for right (as in direction) is 'dexter', and use of this word would have maintained an undesired continuity with previous nomenclature.

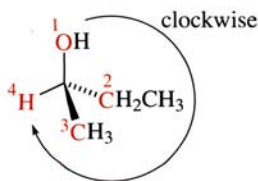
The rules firstly require that the stereogenic centre be specified; next the groups attached to this centre are arranged in a sequence and given numbers 1, 2, 3, 4, corresponding to decreasing atomic number of the atom bonded to the stereogenic carbon. If, in a chiral molecule, two atoms directly bonded to the stereogenic centre are the same, *e.g.* $-\text{CH}_2-\text{CH}_3$ and $-\text{CH}_2-\text{CH}_2\text{Cl}$, the priority sequence is determined by the relative atomic numbers at the first further point along the chains at which a distinction can be made. The convention thus operates in much the same manner as when one looks up a name in a telephone directory. One can, of course, have a chiral molecule in which the priority rankings are the same at all four carbons bonded to the stereogenic carbon, provided that each substituent is distinct by virtue of its characteristics further away from the stereogenic centre.

In the case of a simple molecule, *e.g.* butan-2-ol (**11**), the relevant priorities are assigned as shown. The molecule is then viewed looking toward the atom of lowest priority (H in **11**) along a projection of the H–C bond, through a 'triangle' constructed of atoms 1, 2 and 3. The sense of decreasing priority $1 \rightarrow 2 \rightarrow 3$ is clockwise for **11**, which has *R* configuration. Strictly speaking, the symbol *R* applies to one carbon but in molecules with one stereogenic centre this is often not specified.

Assignments of configuration are easier if the atom of lowest priority, 4, typically H, is shown behind the plane of the paper. In cases in which the atom with priority 4 is shown in the plane of, or in front of, the paper, extra care is required, and it is beneficial to use a model and re-position the molecule so that it can be re-drawn with the atom of priority 4 behind the plane of the paper.

In the *R/S* convention, substituents around a stereogenic centre are given priority numbers (1, 2, 3, 4) in order of decreasing atomic number. Substituent 1 ranks higher than substituent 2, etc.

View the lowest priority substituent, number 4, looking through the stereogenic carbon. If the substituent priority sequence $1 \rightarrow 2 \rightarrow 3$ is clockwise, then the configuration is *R*. If anti-clockwise, the configuration is *S*.



11 (*R*)-butan-2-ol

In a wider context, multiple bonds, such as those in $\text{C}=\text{O}$, $\text{C}=\text{CH}_2$, C_6H_5 , $\text{C}=\text{NH}$ and $\text{C}\equiv\text{N}$ are treated as multiple single bonds and the σ and π bonds are given equal status. These five functional groups become, respectively, as shown in Figure 2.4. Rings are treated similarly; for details see Eliel and Wilen.⁶

Multiple bonds are treated as multiple single bonds.

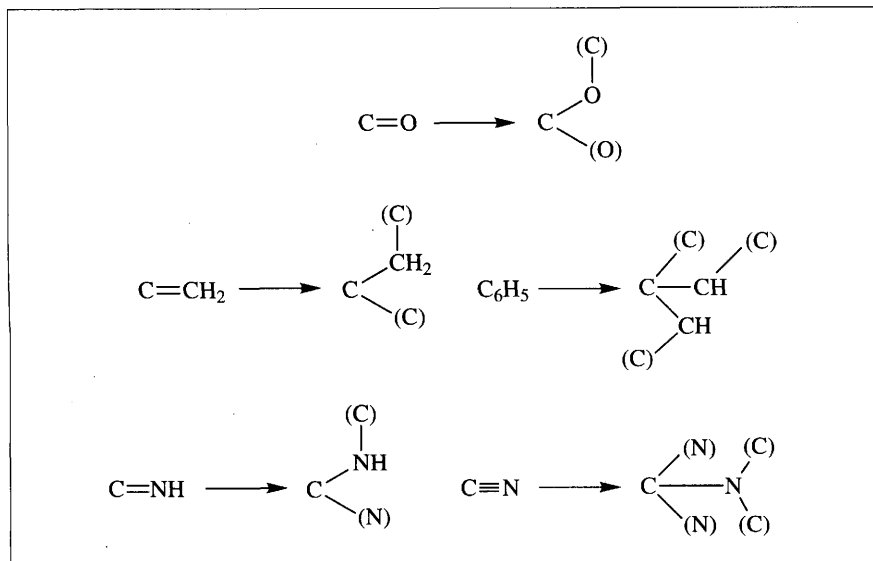


Figure 2.4

Table 2.1 shows the CIP priority sequences for a number of typical organic groups. A compound such as $\text{Cl}_3\text{C}-\text{CH}(\text{OH})-\text{CH}_2\text{Br}$ provides an instructive example on priorities. The group OH clearly has the highest priority, and H has the lowest. The decision between the other two groups cannot be made at the atoms bonded to the stereogenic carbon, as both these are carbon. However, it can be made on the basis of the substituents at these carbons; in particular, one looks for the single highest atomic number (and *not* the aggregate). Therefore, $-\text{CH}_2\text{Br}$ takes precedence over $-\text{CCl}_3$.

For two isotopes, which of course have the same atomic number, priority is assigned on the basis of greater atomic mass, *i.e.* for the isotopes of hydrogen the priority is $\text{T} > \text{D} > \text{H}$. In the case of sulfoxides, for example, the lone pair is a formal substituent, and has the lowest priority of all.

Panico *et al.*⁷ give a survey of the *R/S* convention.

2.3 Enantiomeric Excess, Enantiomeric Ratio

A sample that consists of one enantiomer of a chiral compound is said to be **enantiomerically pure**. This term has mainly replaced 'optically pure' that historically was used because the enantiomer content of a sam-

Table 2.1 CIP sequence rule order for commonly encountered groups (the lower the position of a group in the Table, the greater its priority)

1	H	hydrogen
2	D	deuterium
3	T	tritium
4	CH ₃	methyl
5	CH ₂ CH ₃	ethyl
6	CH ₂ CH ₂ CH ₃	propyl
7	CH ₂ CH ₂ CH ₂ CH ₃	butyl
8	CH ₂ CH ₂ CH(CH ₃) ₂	3-methylbutyl (isopentyl)
9	CH ₂ CH(CH ₃) ₂	2-methylpropyl (isobutyl)
10	CH ₂ CH=CH ₂	prop-2-enyl (allyl)
11	CH ₂ C(CH ₃) ₃	2,2-dimethylpropyl (neopentyl)
12	CH ₂ C≡CH	prop-2-ynyl (propargyl)
13	CH ₂ C ₆ H ₅	phenylmethyl (benzyl)
14	CH(CH ₃) ₂	1-methylethyl (isopropyl)
15	CH=CH ₂	ethenyl (vinyl)
16	CH(CH ₃)CH ₂ CH ₃	1-methylpropyl (s-butyl)
17	C ₆ H ₁₁	cyclohexyl
18	CH=CHCH ₃	prop-1-enyl
19	C(CH ₃) ₃	1,1-dimethylethyl (<i>t</i> -butyl)
20	(CH ₃)C=CH ₂	1-methylethenyl (isopropenyl)
21	C≡CH	ethynyl
22	C ₆ H ₅	phenyl
23	4-CH ₃ C ₆ H ₄	<i>p</i> -tolyl
24	4-NO ₂ C ₆ H ₄	<i>p</i> -nitrophenyl
25	2-CH ₃ C ₆ H ₄	<i>o</i> -tolyl
26	CHO	methanoyl (formyl)
27	COCH ₃	ethanoyl (acetyl)
28	COC ₆ H ₅	benzoyl
29	OCOCH ₃	methoxycarbonyl
30	OCOC(CH ₃) ₃	<i>t</i> -butoxycarbonyl
31	NH ₂	amino
32	*NH ₃	ammonio
33	NHCH ₃	methylamino
34	NHC ₆ H ₅	phenylamino (anilino)
35	NHCOCH ₃	acetylamino
36	NHCOC ₆ H ₅	benzoylamino
37	N(CH ₃) ₂	dimethylamino
38	NO	nitroso
39	NO ₂	nitro
40	OH	hydroxy
41	OCH ₃	methoxy
42	OCH ₂ C ₆ H ₅	benzyloxy
43	OC ₆ H ₅	phenoxy
44	OCOCH ₃	acetoxo
45	F	fluoro
46	SH	mercapto (sulfanyl)
47	SCH ₃	methylsulfanyl (methylthio)
48	SO ₂ CH ₃	methylsulfonyl
49	Cl	chloro
50	SeCH ₃	methylselenanyl
51	Br	bromo
52	I	iodo

ple was always determined polarimetrically. Nowadays, other more accurate methods are employed in many instances (see Chapter 8).

The term '**enantiomeric excess**', which gives a measure of the enantiomeric makeup of a sample that contains enantiomers A and B, is given by:

$$\% \text{ enantiomeric excess} = \frac{\text{no. of moles enantiomer A} - \text{no. of moles of enantiomer B}}{\text{no. of moles of both enantiomers}}$$

The terms in the numerator arise because what the polarimeter registers is a measure of the *difference* in population of the enantiomers in a sample. An enantiomeric excess of 50% signifies that the sample contains 75% of the enantiomer with, say, *R* configuration and 25% of *S*, that is 50% of a 1:1 mixture and 50% of the *R* enantiomer.

Since other techniques (*e.g.* NMR spectroscopy and gas-liquid chromatography) are used to derive enantiomer proportions under conditions in which each enantiomer gives its own 'signal' (Chapter 8), these are now more rationally expressed as a ratio rather than as an excess (this view was projected by Professor D. Seebach, ETH Zurich, at a colloquium at the University of Glasgow, 20 November 1998).

The enantiomeric content of a sample is expressed as an excess (*i.e.* a difference), or better as a ratio.

2.4 Racemization

A **racemic mixture** (or **racemate**) is defined as a 1:1 ratio of enantiomers. This corresponds to an enantiomeric excess of 0%, or an enantiomeric ratio of 1. **Racemization** involves the progressive loss of optical activity with time, usually in accord with a well-defined kinetic process.

In practice, spectra of racemic and enantiomeric forms are indistinguishable, though in principle this will only be true at infinite dilution. In the solid state the difference between a racemate and one enantiomer has features in common with a large box filled with (a) pairs of shoes and (b) left shoes only. Thus the packing in the two instances is not the same. This is reflected in different melting points and densities of the racemate and either enantiomer; these properties are the same for the two enantiomers.

Racemic compounds, or *dl* pairs, arise from (a) deliberate racemization of enantiomers by interconversion (the mechanism is not usually significant), (b) mixing the enantiomers in a 1:1 molar ratio, and (c) synthesis in the absence of a biasing influence that would cause one enantiomer to predominate.

It should be noted that on a molar basis a racemate is more stable than either enantiomer on account of the entropy of mixing (see Alberty and Silbey⁸). The entropy of mixing, $\Delta_{\text{mix}} S$, for the two enantiomers A and B present in equal amounts is given by equation (4):

A racemate is a 1:1 mixture of enantiomers.

$$\Delta_{\text{mix}}S = -nR(x_A \ln x_A + x_B \ln x_B) \quad (4)$$

where n = total number of moles present, and x_A and x_B represent the mole fractions of A and B. The contribution of the entropy of mixing to the free energy of mixing is given by equation (5):

$$T\Delta_{\text{mix}}S = -nRT(x_A \ln x_A + x_B \ln x_B) \quad (5)$$

For $n = 1$, and $x_A = x_B = 0.5$ and a room temperature of 298 K, the free energy of mixing has the value of $T\Delta_{\text{mix}}S = 1.71 \text{ kJ mol}^{-1}$. This is the amount by which a mole of racemate is more stable than a mole of either enantiomer.

It should also be noted:

- (i) that for a molecule with one stereogenic centre it is not possible to have a single molecule of a racemate; one needs at least both a molecule in which the stereogenic centre has *R* configuration and a molecule in which it has *S* (one can alternatively say that one needs both a molecule of *D* configuration and one of *L* configuration).
- (ii) that a racemic sample of a compound is composed of chiral molecules, just as is the case for a sample of either enantiomer.

A term now in use to describe a sample that is partially racemized (or, alternatively, enantiomerically enriched), *i.e.* the sample consists of, say, 70% of enantiomer A and 30% of B, is **scalemic** (from the Greek ‘scalemos’ = lopsided), proposed by Brewster from Purdue University, Indiana (communication to Heathcock *et al.*⁹).

Separation of one or both enantiomers from a racemic mixture is called **resolution**, and it is the converse of racemization. Since resolution usually involves molecules or systems with more than one stereogenic centre, it is dealt with in Chapter 3.

2.5 Homochiral Molecules

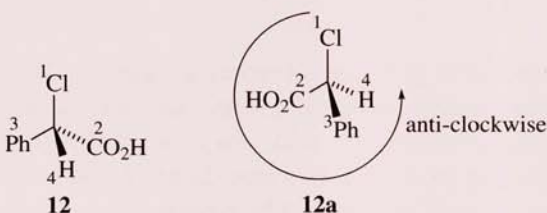
The term **homochiral** was introduced by Kelvin in the 1904 publication of his Baltimore Lecture of 1884 and represented a relationship between molecules (see Mislow¹⁰). Molecules are homochiral if they possess the same sense of chirality. For example, the right hands of a group of people are homochiral (or alike). More recently, and unfortunately, homochiral has been used in the sense of enantiomerically pure, *i.e.* one reads of a ‘homochiral compound’, which clearly violates the original definition. Since some journals permit the latter usage, readers should be aware of the potential for confusion.

To illustrate this, the enzyme hog-kidney acylase hydrolyses the natural enantiomers of *N*-acylamino acids, regardless of the structure of the

R group in $\text{RCH}_2\text{N}(\text{COMe})\text{CO}_2\text{H}$; *N*-acyl-L-amino acids are thus homochirally related (see Mislow and Bickart¹¹).

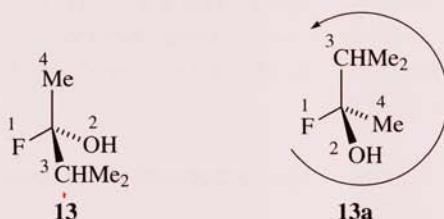
Worked Problems

Q1. Assign configuration to compound **12**.



A. The stereogenic centre is C(2). Priorities of groups attached to the stereogenic centre are $\text{Cl} > \text{CO}_2\text{H} > \text{Ph} > \text{H}$. Rotate structure **12** around the C–Cl bond by 120° in an anti-clockwise direction. The resultant diagram is **12a**, and places the atom of lowest priority behind the page; as no bonds have been broken, there will be no change of configuration. With the eye looking toward, and along, the C(2)–H bond, the priorities decrease $1 > 2 > 3$ in an anticlockwise sense and **12** therefore has *S* configuration. Compound **12** is (*S*)-2-chloro-2-phenylethanoic acid.

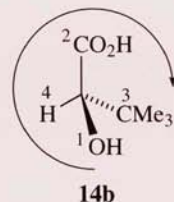
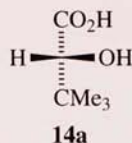
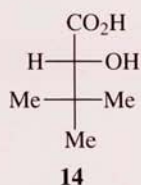
Q2. Assign configuration to compound **13**.



A. The stereogenic centre is C(2). Priorities of groups attached to the stereogenic centre are $\text{F} > \text{OH} > \text{CHMe}_2 > \text{Me}$. If one looks along the C(2)–Me bond from below the molecule, as drawn, one can see that the sequence of priorities decreases $1 > 2 > 3$ in an anti-clockwise sense; hence the configuration is *S*.

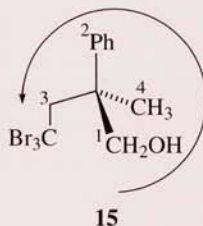
Alternatively, rotate **13** clockwise around the C–F bond by 120° ; this gives **13a**, which has the group of lowest priority behind the page. Again the configuration is *S*. Compound **13** is (*S*)-2-fluoro-3-methylbutan-2-ol.

Q3. Assign configuration to compound **14**.



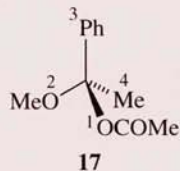
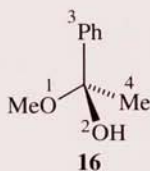
A. Compound **14** is given as a Fischer projection. The lower part is a 1,1-dimethylethyl (*t*-butyl) group, and the compound, 3,3-dimethyl-2-hydroxybutanoic acid, may be re-drawn as in **14a**, which in turn may be shown as stereo diagram **14b** with C(1), C(2) and H in the plane of the page. The configuration is then assigned *R*.

Q4. Assign configuration to compound **15**.



A. Priorities are assigned as shown. Note in particular that the $\text{CH}_2\text{-CBr}_3$ group is given priority 2, even though it contains three bromine atoms, behind CH_2OH and Ph. In this case the decision about priorities is made at the carbons bonded to the stereogenic centre, and the bromines in **15** are too remote to be decisive. The configuration of **15** is *S*.

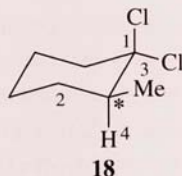
Q5. Assign configurations to compound **16** (a hemiacetal), and its ethanoate (acetate) **17**.



A. With priorities as indicated, **16** has *S* configuration. Although no bonds to the stereogenic centre are broken in conversion of **16** into the ethanoate **17**, the latter now has *R* configuration.

This pair of compounds illustrates a significant point. In assignment of priorities for compound **16**, OMe takes precedence over OH, whereas for **17**, OCOMe now has priority over OMe. Actually, the *absolute* configurations of **16** and **17** are the same, but they are correctly assigned different *R/S* symbols in the CIP convention.

Q6. Assign configuration to compound **18**, 1,1-dichloro-2-methylcyclohexane.



A. Cyclic compounds are treated in the same way as their open-chain analogues, and the priorities are assigned as shown. Note that the numbers in compound **18** refer to the CIP priority assignments around the stereogenic carbon C(2), which is denoted by C*. If, in accordance with the CIP rules, one looks at the C*–H bond from above the molecule as drawn, the configuration is seen to be *S*.

Summary of Key Points

- The principles of polarimetry, which are fundamental for the detection and characterization of chiral molecules, were outlined for molecules that contain a single stereogenic centre. Polarimetric measurements are most commonly carried out at 589.1 nm, the wavelength of the sodium D line. Specific rotation was defined.
- Two conventions that are very useful in defining the stereochemistry of molecules were described.
- The D/L convention is used for sugars and amino acids. Sugars are related to a standard, D-(+)-glyceraldehyde, which has one stereogenic centre.
- Naturally occurring amino acids are also related to L-(–)-glyceraldehyde by chemical transformations such as those indicated briefly for serine, which involves L-lactic acid as a relay.
- The *R/S* or CIP (Cahn–Ingold–Prelog) convention involves assignment of priorities, based on atomic numbers, to the four substituents at a stereogenic carbon or centre. This latter con-

3

Molecules with Two (or More) Stereogenic Centres

Aims

By the end of this chapter you should be familiar with:

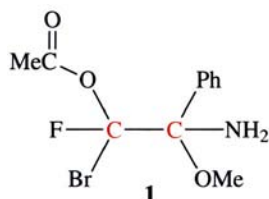
- Diastereoisomers in molecules with two or more stereogenic centres
- *meso* configurations when the substituents at two stereogenic centres are identical
- The terms *erythro* and *threo*, and *syn* and *anti*, when applied to molecules with two stereogenic centres
- Use of the terms ‘epimers’, *exo* and *endo* in connection with nomenclature of bicyclic molecules
- Separation of racemic compounds into enantiomers by resolutions that involve:
 - Recrystallization of derived diastereoisomers
 - Chiral chromatography
 - Use of enzymes that react exclusively with one enantiomer

3.1 Enantiomers and Diastereoisomers

In the previous chapter it was shown that a molecule with one stereogenic centre gives rise to two enantiomers. A molecule with two stereogenic centres gives rise to a *maximum* of four stereoisomers; a molecule with three stereogenic centres can have a maximum of eight stereoisomers, and so on. In general, if a molecule contains n stereogenic centres, a maximum of 2^n stereoisomers can exist.

The molecule **1** has two stereogenic centres and has a different set of substituents at each centre; accordingly, there are four stereoisomers of **1**. Four stereoisomers also result if in **1** one replaces, say, an NH_2 by an

A molecule with n stereogenic centres can have a maximum of 2^n stereoisomers.



F, so that one substituent, *i.e.* F, is now common to both stereogenic centres.

In designating the stereochemistry of, say, **1**, the molecule is first numbered systematically. The configuration at each stereogenic centre is determined in the same way as for molecules with one stereogenic centre (Chapter 2). The symbol *R* or *S* is then associated with a particular carbon by its number and the pair of symbols is placed in brackets. Accordingly for **1** the four stereoisomers can be (1*R*,2*R*), (1*R*,2*S*), (1*S*,2*S*) and (1*S*,2*R*), and these are shown schematically in Figure 3.1.

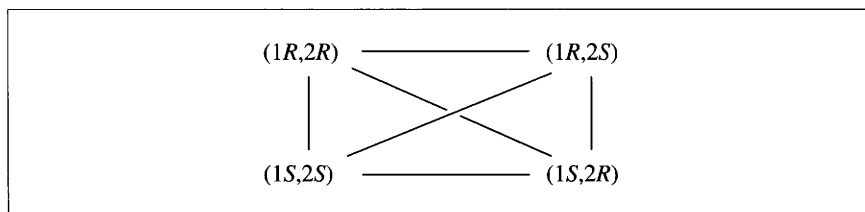


Figure 3.1

Diastereoisomers have different physical properties and react at different rates: they are stereoisomers that are not enantiomers.

A particular stereoisomer of **1** with two stereogenic centres can have only one enantiomer, but more than one diastereoisomer.

It can be seen that the stereoisomers with configuration (1*R*,2*R*) and (1*S*,2*S*) are related as enantiomers, and the (1*R*,2*S*) and (1*S*,2*R*) stereoisomers are also related as enantiomers. What then is the relationship between the stereoisomers of **1** with configurations (1*R*,2*S*) and (1*R*,2*R*), for example? With respect to each other, these have one stereogenic centre of common, (1*R*), configuration, and the other of opposite, (2*S*) and (2*R*), configuration. Clearly these stereoisomers are not enantiomers. Instead, they are related as **diastereoisomers** and the same can be said for the (1*S*,2*R*) and (1*S*,2*S*) stereoisomers. The most concise definition, given in the plural, is: 'diastereoisomers are stereoisomers that are not enantiomers'.

For a compound that has two stereogenic centres, and has different groups attached to each such centre as in **1**, the representation in Figure 1 shows that molecules related vertically are enantiomers, whereas those related either diagonally or horizontally are diastereoisomers. The interrelationships in Figure 1 show that whereas a given chiral molecule can possess only one enantiomer, it can have more than one diastereoisomer.

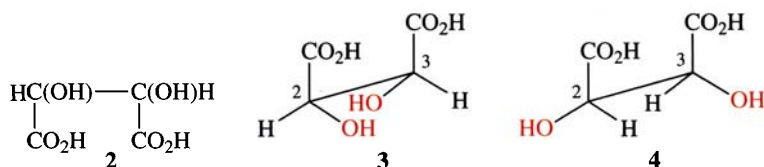
Molecules that are related as diastereoisomers have different melting points and solubilities, and have different rates of reaction. In some cases, these different rates of reaction can lead to a particular product being formed from a reaction of one diastereoisomer but not the other. Use is made of differences of, for example, solubility of diastereoisomers in the resolution of a racemic mixture (Section 3.6).

3.2 Meso Configuration

In contrast to the general case of **1**, we now consider molecules with two stereogenic centres, but which carry an identical set of substituents at each carbon. We then analyse the relative configurations of these substituents and the stereochemical consequences.

As a prototype molecule, tartaric acid (**2**; 2,3-dihydroxybutanedioic acid), shown here without stereochemical specification, is chosen. Tartaric acid is a compound of considerable historical importance (Section 3.6), and derivatives of tartaric acid are still compounds of contemporary research interest (see Seebach *et al.*¹). (2*R*,3*R*)-(+)-Tartaric acid is shown in sawhorse projection in **3**.

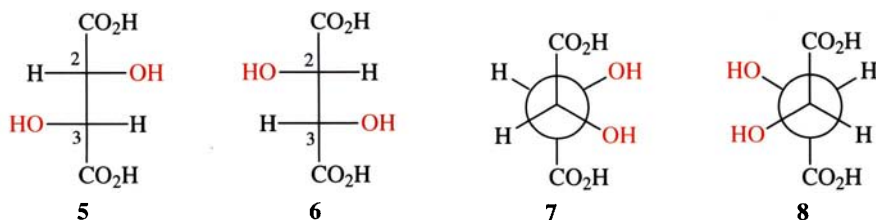
A molecule with two stereogenic centres, each of which carries the same substituents ($C_{abc}-C_{abc}$), e.g. tartaric acid, is limited to a total of three rather than four stereoisomers.



With the aid of molecular models verify that the configurations of C(2) and C(3) are correct.

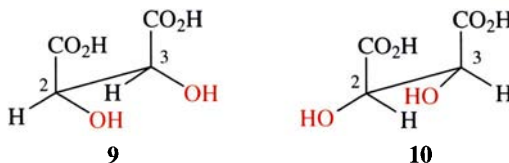
The enantiomer of **3** is shown in **4** and is (2*S*,3*S*)-(–)-tartaric acid. Of course, in principle any conformation of **3** may be used, with its mirror image, to demonstrate that the compounds in the pair are enantiomers; however, use of the eclipsed forms **3** and **4** is visually easier.

Fischer projections of **3** and **4** are given in **5** and **6**, respectively, and the corresponding Newman projections, looking along the C(2)–C(3) bond, are shown in **7** and **8**, but now with the rotation of the ‘back’ groups around the C(2)–C(3) bond such that the molecules are shown in staggered and visually less cramped conformations.



We now consider the other configurations of tartaric acid, *i.e.* (2*R*,3*S*) **9** and (2*S*,3*R*) **10**. Here, the presence of identical sets of substituents at C(2) and C(3) brings about a difference from the general case which was shown by **1**. Inspection of **9** and **10** reveals that **10** is a mirror image of **9** (and *vice versa*). However, **9** and **10** are identical and superimposable.

The *meso* stereoisomer of tartaric acid is achiral, and possesses two self-cancelling stereogenic centres of opposite configuration.



Use molecular models to check that this is so.

Therefore, and importantly, this molecule is achiral despite containing two stereogenic centres. This can be shown in two ways.

(a) If one focuses on **9**, in the conformation shown, there is seen to be a plane of symmetry in this molecule half-way along the C(2)–C(3) bond, such that $\text{--CO}_2\text{H}$ reflects on to $\text{--CO}_2\text{H}$, --OH on to --OH , and --H on to --H .

(b) Take a molecular model of **9**, and break the C(2)–C(3) bond in half, and then mark the ‘break’ on both sides with tape, or similar, and regard the taped ends as identical substituents. Examination of the resultant taped fragments clearly reveals that C(2) and C(3) are of opposite configuration.

In Chapter 2, a number of everyday objects were considered; some are chiral and some are not. Those that are not chiral have a plane of symmetry. The same characteristics operate at a molecular level and one can therefore state that if a molecule possesses a plane of symmetry it is not chiral. It is possible to see this for simple molecules such as methane and dichloromethane and also for *meso*-tartaric acid **9**, drawn so that the plane of symmetry may readily be recognized. In assessment of whether a molecule is, or is not, chiral it is advisable to check for both (a) the presence of stereogenic centres and (b) the presence of a plane of symmetry.

As an exercise, take a model of **9**, as drawn, and rotate the rear groups clockwise by 60° around the C(2)–C(3) bond. Then confirm that in this conformation the molecule is now chiral.

However, **9** is *not* chiral and one can accordingly amplify the definition above to state: if a molecule possesses a plane of symmetry *in any energetically available conformation*, it cannot be chiral. This statement holds whatever the complexity of the molecule. However, it should be noted that chiral molecules can possess axes of symmetry, and such molecules are considered in Chapter 5.

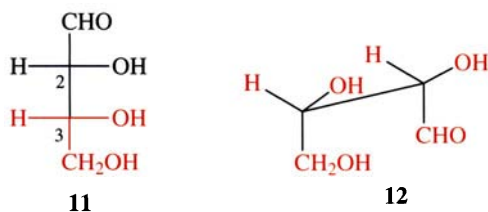
The achiral nature of **9** is also implicit in the designation of configurations ($2R,3S$) or ($2S,3R$) with identical sets of substituents at the stereogenic centres. As regards optical activity, the effects of C(2) and C(3) in **9** are self-cancelling because the stereogenic centres have opposite configuration; indeed, in early literature compounds such as **9** were described as ‘internally compensated’. Nowadays, such compounds are given the prefix *meso*, and use of the term is general; *meso* compounds are always achiral (and optically inactive) even though they have two or more stereogenic carbons.

Confirm that **3** and **4** are enantiomers, and that **3** and **9**, also **4** and **9**, are diastereoisomers.

3.3 Erythro/Threo and Syn/Anti Configurations

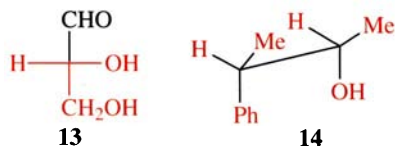
Let us now look at molecules with two stereogenic centres, usually adjacent, which have two common substituents. The names of the two configurations are taken from two carbohydrates, but are now used quite generally. D-Erythrose shown in Fischer **11** and sawhorse **12** projections in its open-chain forms is such a compound. In **12** the pairs of like substituents are shown in an eclipsed conformation.

A molecule with two stereogenic centres with two substituents in common ($C_{abc}-C_{abd}$) gives rise to *erythro* and *threo* diastereoisomers.

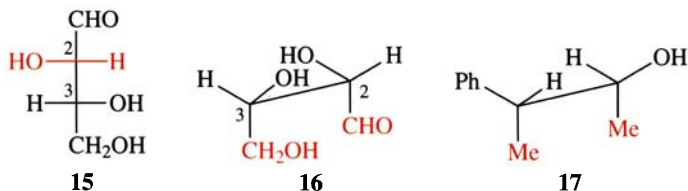


Firstly, one might ask: why is **11** given the configuration D? This assignment is based on the configuration of the stereogenic carbon that is furthest from the carbonyl group [*i.e.* C(3) in **11**] with respect to that of D-glyceraldehyde (**13**). Since they are the same, **11** is given D configuration. This protocol can also be applied to longer chain sugars such as pentoses and hexoses (erythrose and threose are themselves tetroses). If, as in **12**, it is possible to eclipse each of the like substituents, H with H, OH with OH, and also the ‘unlike’ substituents CHO and CH₂OH, the configuration is *erythro*. The nomenclature is quite general; thus *erythro*-3-phenylbutan-2-ol is shown in **14**. Of course, **12** and **14**, and indeed all *erythro* compounds, have enantiomers, but this does not concern us here.

The second configuration is termed *threo*, and originates from the sugar ‘threose’; D-threose is shown in Fischer **15** and sawhorse **16** projections. With respect to D-erythrose (**11**), only the configuration at C(2)



has changed. Two consequences follow from this: (i) **11** and **15** are related as diastereoisomers and (ii) the stereochemical symbol is D in both **11** and **15**.



If a molecule ($C_{abc}-C_{abd}$) in an *erythro* configuration is converted into one in which the unlike substituents, say c and d, are each transformed into a common substituent, say e, i.e. to give ($C_{abe}-C_{abe}$), the resultant compound is *meso*.

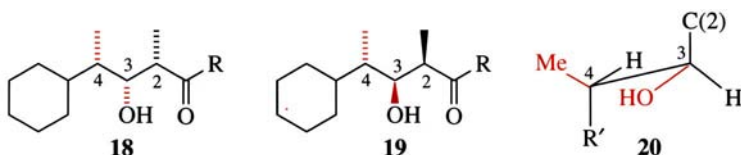
The configurations at C(2) and C(3) in D-threose are now such that in sawhorse projections one can eclipse only (a) the H atoms, or (b) the OH groups or (c) the unlike substituents, CH_2OH and CHO, at any one time. Again the nomenclature is quite general, and one enantiomer of *threo*-3-phenylbutan-2-ol is shown in **17**.

An *aide-memoire* to assist in assignment of *erythro* and *threo* configurations is this: if one takes erythrose (**11**), the product obtained after oxidation of both the CH_2OH and CHO groups to $COOH$ is *meso*-tartaric acid (**9**).

Use of the terms *erythro* and *threo* is quite explicit in the context of two adjacent stereogenic centres, each of which has two sets of substituents in common. However, some confusion has arisen when the *erythro*/*threo* nomenclature has been applied to systems in which there is only one common substituent on the two adjacent carbons. Such a case is found in the products of asymmetric aldol reactions. The mechanism of this reaction lies outside the scope of this chapter; however, we give here the basis of an alternative nomenclature system, which is now becoming more common for products of this reaction, and which was introduced by Masamune *et al.*² The main chain is drawn as a zig-zag, and lies in the plane of the paper. Substituents that lie on the same side of the plane are termed *syn*, and those that do not are termed *anti*. Thus in the case of **18** the 3,4 stereochemistry is *syn* and in **19** it is *anti*; **18** can be fully described as *syn,syn* and **19** as *syn,anti*. The drawings here represent essentially a telescoping of sawhorse projections, and the position of the eye has moved so as to view the molecule from a more 'side-ways on' aspect. The wedges imply that the bond comes out from the paper toward the viewer and the 'hatched' (or broken) lines imply that the bond is directed behind the paper. Although sawhorse projections are ideal for inspection of the configurations around a two-carbon unit,

A *syn/anti* notation for relative configuration at sp^3 carbons is described.

they become unwieldy when chains of more than two carbons are involved, and in such cases stereo drawings are now preferred. A saw-horse projection of **19** along the C(4)–C(3) bond is given in **20** ($R' = \text{cyclohexyl}$).



As an alternative to *erythro* and *threo*, two further conventions have been introduced for the specification of relative configuration in molecules that contain two (or more) stereogenic centres. In a molecule such as (2*R*,3*R*)-tartaric acid (**3**) or (2*S*,3*S*)-tartaric acid (**4**), the configurations are described as *l* for like. *meso*-Tartaric acid (**9**) has configuration (2*R*,3*S*) [or (2*S*,3*R*)] and is described as *u* for unlike. Compounds with more than two stereogenic centres, *e.g.* (1*R*,2*R*,3*S*), have relative configurations specified by a comparison of their configurations in sequence. A compound with the above configuration, and its enantiomer, can be collectively described as *l.u.* (see Seebach and Prelog³).

Alternatively, in a convention employed by *Chemical Abstracts*, the two enantiomers (2*R*,3*R*)- and (2*S*,3*S*)-tartaric acid can be specified by (2*R**,3*R**). The symbols (2*R**,3*R**) indicate that if C(2) has *R* configuration, so does C(3), and likewise that if C(2) has *S* configuration, so does C(3). *meso*-Tartaric acid can formally be described as having (2*R**,3*S**) configuration and this indicates that if C(2) has *R* configuration, C(3) has the *S* and *vice versa*.

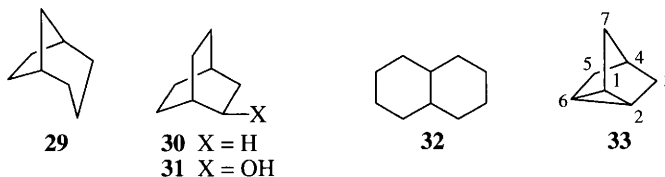
Note that whereas *R* and *S* symbols apply to specific carbons, *D* and *L* apply to particular *reference* carbons, as has been described for sugars in this section. In the case of amino acids the configuration, *D* or *L*, is always taken from the α carbon, which is bonded to both NH_2 and CO_2H . This applies even when there is more than one stereogenic centre in the amino acid, as in the case of, for example, threonine. Further discussion of this point lies outside the scope of this text but is dealt with explicitly in the text by Loudon.⁴

3.4 Caged Compounds with Two Stereogenic Bridgehead Carbons

An example of a compound with two stereogenic carbons in a rigid caged structure is camphor, a naturally occurring bicyclic ketone which is more abundant in the (+)-form **21**. The presence of the carbonyl group in camphor results in two stereogenic centres, at C(1) and C(4).

A bicyclic compound, chiral because of stereogenic bridgehead carbons, is restricted to two enantiomers, and no diastereoisomers.

commonly known as decalin. The case of an additional ring is exemplified by the C_7H_{10} compound nortricyclene (**33**). The systematic name for **33** is tricyclo[2.2.1.0^{2,6}]heptane, the last entry within the bracket signifies that, compared to **28**, an additional zero-carbon bridge has been introduced between C(2) and C(6) to form the three-membered ring.



There are a couple of further points, one of which is exemplified by camphor (**21**) and the borneols. These molecules have the functional groups on one of two two-carbon bridges. Accordingly, the numbering is arranged to be as low as possible, and so in camphor, for example, the carbonyl carbon is C(2).

If a substituent on the smallest bridge is directed toward the next smallest bridge, it is termed *syn*; if the substituent points toward the largest bridge, it is termed *anti*. However, camphor, for example, has two bridges that both contain two carbons, *viz.* C(2)–C(3) and C(5)–C(6). Now if the substituent on the smallest bridge points toward the bridge that itself carries substituents [in camphor C(2) is part of the carbonyl group], it is called *syn*. Where the substituent on the smallest bridge in camphor points toward the unsubstituted bridge, it is called *anti*. In camphor (**21**) the methyl groups containing C(8) and C(9) are therefore labelled *syn* and *anti*, respectively.

3.6 Separation of Enantiomers: Resolution

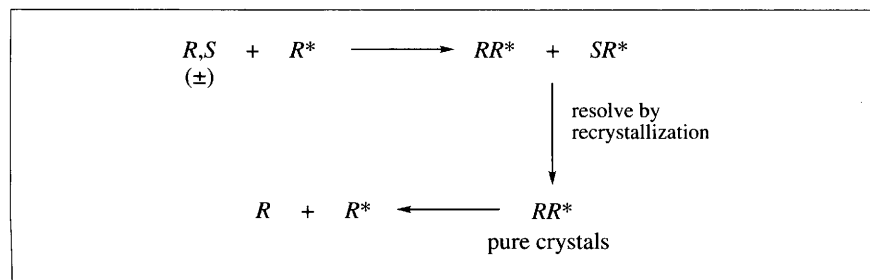
Resolution can be thought of as the converse of racemization (Section 2.4). One starts with a 50:50 mixture of both enantiomers and separates this mixture into the individual enantiomers. Of course, for some purposes one may only want one enantiomer, and recovery of the second enantiomer can be painstaking. Since enantiomers have identical properties, including solubility, separation of enantiomers by recrystallization is quite rare. It was, however, such a crystallization by Pasteur in 1848 that opened up the field of resolution. Pasteur's key observation was that two distinct but related types of crystal were obtained from an aqueous solution of the sodium ammonium salt of racemic tartaric acid. The two types of crystal were related as object and non-superimposable mirror image, and one type was identical to the dextrorotatory crystals of sodium ammonium tartrate obtained from (+)-tartaric acid, itself obtained as a by-product of wine-making.

Pasteur separated his crystals manually with the aid of a magnifying glass and tweezers. Moreover, he demonstrated that solutions of the two types of crystals had the same value of specific rotation, though of opposite sign. In correlating the macroscopic with the molecular, the important and fundamental conclusion was drawn that the molecules from which the two crystals were composed were likewise non-superimposable mirror images. Like many important discoveries, Pasteur's work was a combination of perceptiveness, tenacity and happenstance, the last because he worked with the only known tartrate salt that gives enantiomeric crystals with different crystal forms. Additionally, these different crystals form only at temperatures below 25 °C. From a knowledge that he worked in Paris, one can conclude that Pasteur did not carry out his experiments at the height of summer.

Resolution by recrystallization depends on conversion of enantiomers into diastereoisomers, usually salts, and then exploiting solubility differences.

Typically, resolution depends on the conversion of enantiomers, which possess identical physical properties, into diastereoisomers, which do not, and then exploiting the difference in physical properties in order to separate the diastereoisomers. Finally, the diastereoisomers are reconverted to the component, and now separated, enantiomers.

In schematic terms the separation of racemic compound R,S into enantiomers by reaction with a single enantiomer of R^* is represented in Scheme 3.1. RR^* and SR^* are diastereomers, and can be covalent compounds or salts. R^* is chosen so that the solubilities of RR^* and SR^* in a particular solvent are different. The racemic mixture, now in combination with R^* , i.e. RR^* and SR^* , is now recrystallized and one diastereoisomer, say RR^* , will crystallize out whereas the other, SR^* , will stay in solution. Sometimes when the first crop of crystals is merely enriched in RR^* , a number of further recrystallizations will be necessary. Salts are the preferred diastereoisomers (RR^* and SR^*), because they are both easily formed and readily re-converted to the starting enantiomers, following separation.

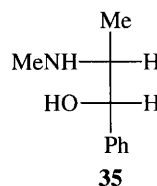
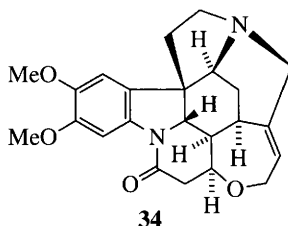


Scheme 3.1 Resolution of a racemic compound R,S to give the more insoluble RR^* as crystals, whereas the diastereoisomer SR^* remains in solution

Resolution of a carboxylic acid can be achieved directly by formation of diastereomeric salts with an amine. Naturally occurring bases such as brucine (34) or ephedrine (35) are often used, though it should be mentioned that choice of a suitable amine is usually a matter of trial and

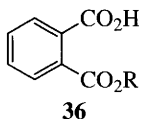
Resolution of alcohols usually involves synthesis of a phthalate half-ester, then formation of a diastereomeric salt.

error and is probably related to crystal packing of the diastereoisomeric salts. In common with camphor, brucine, a naturally occurring alkaloid, has its stereogenic centres in fused ring systems; nature makes only one enantiomer, (–)-brucine (**34**).



How many stereogenic centres are there in brucine?

Resolution of alcohols requires a more ingenious approach in order to acquire suitable diastereomeric salts for resolution by recrystallization. The racemic alcohol, ROH, to be resolved is first converted to its phthalate monoester **36** (phthalic acid is benzene-1,2-dicarboxylic acid). The strategy here is to leave one carboxylic acid group free, and available to form diastereoisomeric salts with an amine such as **34** or **35**. Once the appropriate ammonium salt of **36** has been resolved, the amine is removed by acidification to give one enantiomer of **36**, and this ester is then hydrolysed with alkali to give one enantiomer of ROH. The more soluble diastereomeric ammonium salt can then, in principle, be processed similarly to yield the other enantiomer.



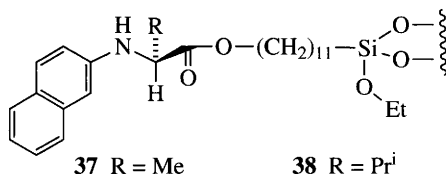
The success of chiral column chromatography depends on diastereoisomeric interactions between the chiral stationary phase and the enantiomers; this leads to differential adsorption of enantiomers.

3.7 Separation of Enantiomers by Chromatography

It is also possible to resolve enantiomers by means of column chromatography. If a column consists of, or contains, a suitable chiral stationary phase, the enantiomers should be eluted, *i.e.* washed from the column, at different rates. This is because, although there is no formation of chemical bonds, the enantiomers now experience diastereomeric interactions with the chiral stationary phase. If there is a high degree of 'chiral recognition' of one particular enantiomer by the stationary phase, then it will be preferentially adsorbed on the chiral stationary phase and accordingly eluted more slowly. Good separations of quite sizeable quantities (50 g per elution) can be achieved in some cases. Column chromatography is described in the books by Harwood *et al.*⁵ and Armarego and Perrin;⁶ chiral chromatography is the subject of a paper by Pirkle *et al.*⁷ see also Lochmüller.⁸

Chiral stationary phases can consist of starch, which for instance

allows almost complete resolution of mandelic acid (2-hydroxy-2-phenylethanoic acid), $\text{PhCH(OH)CO}_2\text{H}$. Synthetic stationary phases such as **37**, derived from the enantiomerically pure amino acid alanine, and **38**, likewise from valine, are effective in resolution of alcohols, amines and amino acids (both α and β), with the proviso that these compounds first be derivatized, *e.g.* an alcohol must be transformed into an ester before being passed down the column.



3.8 Resolution with Enzymes

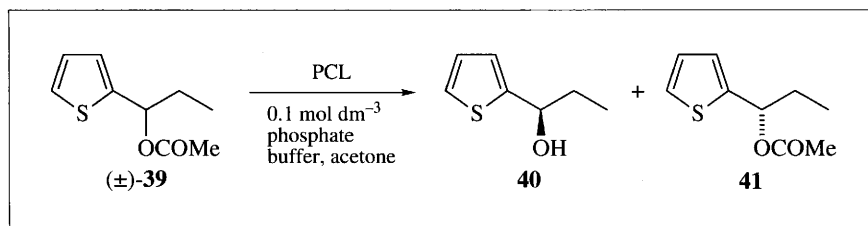
Enzymes are compounds of high molar mass, generally contain many amide groups, and catalyse certain reactions in living cells. All enzymes are catalysts; they are themselves chiral, and in their natural aqueous environment are very stereoselective. Some enzymes are highly selective in the reactions that they catalyse, whereas others are more broadly receptive, and in general enzyme-catalysed reactions tend to be rapid. A discussion of the characteristics of enzymes and their mode of action is given by Stryer.⁹ Furthermore, enzymes have been instrumental in bringing about resolution of enantiomers under laboratory conditions. A quantitative analysis of biochemical kinetic resolutions of enantiomers has been made by Chen *et al.*¹⁰

In order to give a flavour of the method, three examples are presented here.

As part of a project to synthesize one enantiomer of a chiral insect pheromone, resolution of **39** was required. This was achieved by enantioselective hydrolysis of racemic **39** to produce alcohol **40** (Scheme 3.2).

A suitable enzyme catalyses enantioselective hydrolysis of ester **39**. This gives an alcohol of one configuration and leaves behind the ester of the other. Differentiation between enantiomers by enzymes is almost total.

Verify that this alcohol has *R* configuration.

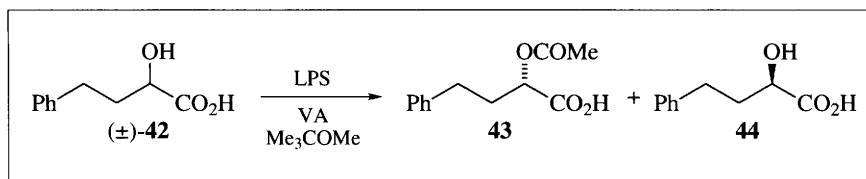


Scheme 3.2

The other enantiomer was left untouched by *Pseudomonas cepacia* lipase (PCL), the enzyme of choice. After reaction is complete, the resolved alcohol must be separated from its enantiomeric acetate **41**. It is noteworthy that the enzyme functions in a buffered aqueous acetone medium; acetone does not interfere with the action of the enzyme.

There is now a sufficient data bank of enzymic resolutions for these to guide the choice of enzyme for a new resolution. Although at 18 °C and after 60% conversion it was possible to obtain acetate **41** of 99% enantiomeric excess (*ee*), it should be recognized that use of enzymes is not always the complete answer. In the above resolution, smaller *ee* values were obtained at higher temperatures and at larger conversions, findings that are perhaps not surprising if one considers that the enzymes are here operating outside their natural environment (see Kang *et al.*¹¹).

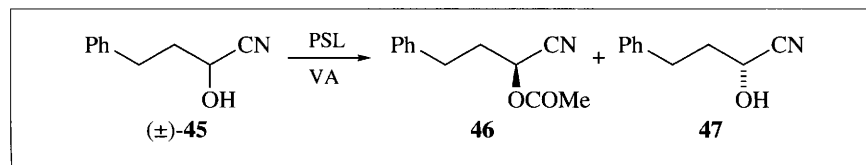
In a second investigation, racemic 2-hydroxy-4-phenylbutanoic acid (**42**) was reacted with lipase PS (LPS) and vinyl acetate (VA) (ethenyl ethanoate) in *t*-butyl methyl ether (2-methoxy-2-methylpropane) to give acetate (*S*)-**43** in 35% yield and in 99% *ee*; unreacted alcohol **44** was found to have 99% *ee* with *R* configuration predominant (Scheme 3.3).



Scheme 3.3

In this resolution involving transesterification, the enzyme is catalysing the acetylation by vinyl acetate of the hydroxyl group of **42** with *S* configuration. This use of vinyl acetate is now fairly common in enzymatic resolution (see Chadha and Manohar¹²); hexanoic anhydride is also sometimes used in this context.

Thirdly, the lipoprotein lipase from *Pseudomonas* species (PSL) reacted with racemic cyanohydrin **45** to give the acetate (ethanoate) **46** of *S* configuration with 98% *ee* after 59% conversion (Scheme 3.4). The unreacted alcohol, effectively now resolved into the enantiomer of *R* configuration **47**, because of its rejection by the enzyme, was then converted into the hydroxy ester **48**, which is an important intermediate in the synthesis of an angiotensin-converting enzyme inhibitor (see Wang *et al.*¹³).

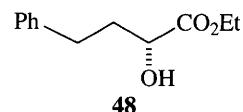


Scheme 3.4

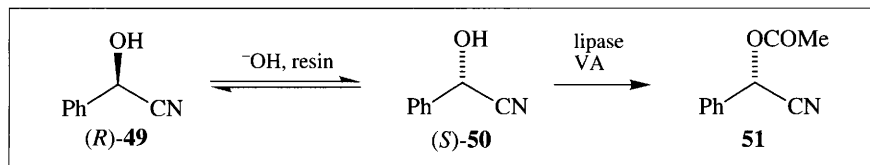
In the presence of vinyl acetate as acylating agent, racemic hydroxy acid **42** was converted enzymatically into the (*S*)-acetate; the (*R*)-hydroxy acid was unaffected.

Enzymic resolutions involve acceptance by the enzyme, which is a very finely honed chiral system, of one enantiomer of a racemic compound, but not the other. The selective acceptance arises because interactions between the enzyme and the enantiomers are diastereomeric. In its natural environment, the ability of an enzyme to discriminate between enantiomers is virtually absolute. In addition to their stereoselectivity, some enzymes can react at very high rates. Each round of catalysis by the enzyme carbonic anhydrase with its physiological substrate occurs in about 1.7 μ s at room temperature, although for a small number of other enzymes, best exemplified by the more lethargic lysozyme, the corresponding figure is about a million times slower. Accordingly, the enzyme-catalysed hydrolysis of, say, one enantiomer of an ester proceeds at a finite rate and hydrolysis of the other not at all. Resolutions such as those of **39**, **42** and **45** therefore have a kinetic basis and are also known as kinetic resolutions.

It should be evident that the maximum yield of a particular enantiomer normally available from a racemic mixture is 50%. However, in some enzymic catalysed kinetic resolutions it is possible to obtain >50% yield of one enantiomer from a racemate. For this to occur, it is necessary to have the desired chemical reaction, *e.g.* enzyme-catalysed stereoselective esterification, occurring at the same time as the enantiomers of the racemic starting compound are interconverting under equilibrium conditions. A successful example of this technique is provided by benzaldehyde cyanhydrin (2-hydroxy-2-phenylacetonitrile), whose *R* and *S* enantiomers, **49** and **50**, respectively, equilibrate in the presence of a basic anion-exchange resin (Scheme 3.5). In the presence of lipase, (*S*)-benzaldehyde cyanhydrin acetate **51** was formed in 95% yield and in 84% enantiomeric excess (see Inagaki *et al.*¹⁴ and Ward¹⁵).



Yields of >50% of one enantiomer may be obtained from a racemic compound if, under the conditions of the enantioselective reaction, the enantiomers of the starting material are interconverting.



Scheme 3.5

As an alternative to resolution, one can start with an enantiomerically pure compound that occurs naturally, and the most common examples of these are amino acids, sugars and terpenes. This group of compounds is known collectively as the 'chiral pool'. The procedure then is to transform the compound of choice from the above group into the desired product by chemical synthesis, with care taken to avoid racemization at stereogenic centres.

5

Chirality without Stereogenic Carbon

Aims

By the end of this chapter you should be familiar with:

- Appropriately substituted allenes and biphenyls as examples of compounds without a stereogenic centre
- The absolute configurations of chiral allenes and biphenyls
- Formation of chiral optically active compounds in which the stereogenic centre can be, for example, Si, Ge, Sn, N, P, As, S or Se

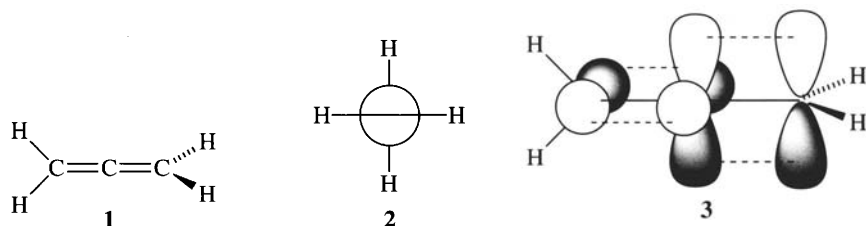
5.1 Allenes and Related Molecules

Nearly all the examples of chiral molecules so far encountered possess one or more stereogenic carbons. In this chapter the scope is widened, and organic molecules are considered that are chiral but do not have stereogenic centres, as also are some molecules in which the stereogenic centre is not carbon. In all cases the condition for chirality, namely that a molecule and its mirror image are not superimposable, is met.

The first class to be considered is the **allenes**. Allene (propa-1,2-diene) itself is represented in **1**, and as a Newman projection in **2**; allene is isomeric with propyne. The two hydrogens at each terminal carbon have a bond angle, HCH, of 120° , and of course these three atoms lie in a plane. However, in allene the two HCH planes are at right angles to each other, and are said to lie in **orthogonal** planes. This is apparent from the Newman projection **2**. It follows that the two π bonds are also orthogonal, and accordingly there is no conjugation between the π bonds. A representation of the bonding in allene is given in **3**; note that the central carbon is sp hybridized. Allenes are not alone in having π bonding of this type; other examples include ketenes, $R_2C=C=O$, isocyanates,

In allenes (a) the adjacent π bonds are orthogonal and (b) each terminal carbon and its two substituents lie in a plane, and these planes are orthogonal.

$\text{RN}=\text{C}=\text{O}$, diimides, $\text{RN}=\text{C}=\text{NR}$, the nitronium ion, $[\text{O}=\text{N}=\text{O}]^+$ and, most commonly, carbon dioxide, $\text{O}=\text{C}=\text{O}$.

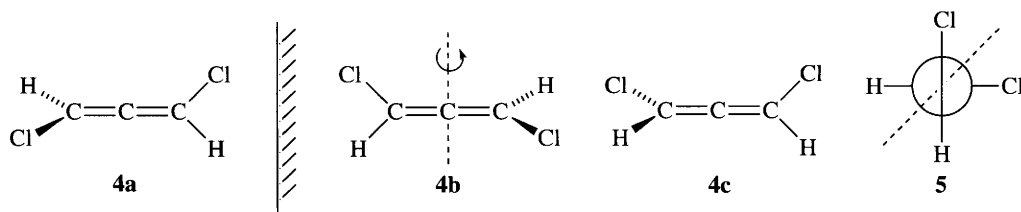


The prediction that certain allenes could exist as enantiomers was made by van't Hoff in 1875; Maitland and Mills finally verified his prediction (in 1935/36) after intense effort in many laboratories.

We consider the example of 1,3-dichloropropa-1,2-diene, which is shown with its mirror image in **4a** and **4b**. These two structures are related as enantiomers.

If possible make a model to verify this, though not all model kits will have the facility to represent the central carbon. Take structure **4b** and rotate it by 180° around a vertical axis through the central carbon; this gives **4c**, and neither by this, nor by any other manoeuvre, can **4b** be superimposed on **4a**.

Enantiomers of chiral allenes are well documented and these molecules are not prone to racemization.



Although we have shown that **4** is chiral, and that it has no stereogenic centre, it is not without symmetry. This is best seen with the aid of a Newman projection of **4**, which is shown in **5**. A two-fold (C_2) axis of symmetry exists as shown; this passes through the central carbon, $\text{C}(2)$, and bisects the right angle between the two chlorine atoms (and likewise the two hydrogen atoms). Rotation about this axis by 180° gives an identical molecule. Because of the axis of symmetry, **4** cannot be said to be asymmetric, and similar situations are found in certain other chiral molecules. This has had implications for stereochemical nomenclature. In particular, to avoid confusion the term 'asymmetric carbon' is now little used even for an sp^3 hybridized carbon that carries four

Allenes with the same two substituents at each terminal carbon are chiral, can be resolved, and are optically stable. Allenes with three or four different substituents are also chiral. These molecules have no stereogenic centre.

Alkylidenecyclohexanes and cyclohexanone oximes, substituted as in **6** and **8**, respectively, have no stereogenic centre and are also chiral.

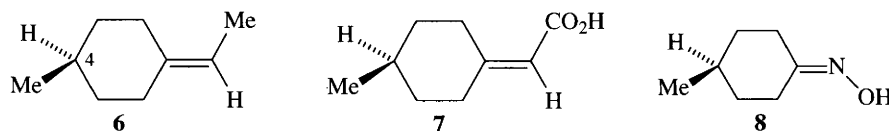
different substituents, and the term has been superseded by ‘**stereogenic carbon**’. The phrase ‘**asymmetric synthesis**’ survives, however. The demonstration that chirality can exist in a *molecule with an axis of symmetry*, e.g. **4**, is in direct contrast to the case of a *molecule with a plane of symmetry* (Chapter 3), in which chirality is not possible.

In order for an allene to be chiral a *minimum* of two different groups, both of which must be present at *each* terminal carbon, as in **4**, is required. This situation contrasts with the four different atoms or groups that must be bonded to an sp^3 hybridized carbon to create a chiral molecule. However, chiral allenes may, of course, carry three, or four, different groups, and enantiomers of such examples are known.

Compounds such as the alkylidenecyclohexane **6** are distinct from, but related to, chiral allenes. Here, one of the double bonds of the allene is replaced by a six-membered ring in which the substituents at the 4-position are the same as those at the terminal vinyl (ethenyl) carbon.

With the aid of a molecular model, verify that the resultant molecule **6** is chiral and note that the pattern of substituents in **6** is reminiscent of that in **4**, and that again there is no stereogenic carbon.

Compound **6**, as with disubstituted allenes such as **4**, illustrates the minimum requirements for chirality, but alkylidenecyclohexanes are known in which four different substituents occupy the key positions. Interestingly, the carboxylic acid **7** was the first example of this type of compound to be resolved into enantiomers, and this feat was achieved over 25 years before the resolution of an allene. One can, of course, consider other than six-membered rings in the context of chirality without stereogenic centres, and examples of alkylidenecyclobutanes are well documented (see Rossi and Diversi¹ and Runge²).



We have previously seen how geometric isomerism can exist around both the carbon–carbon double bond in an alkene and the carbon–nitrogen double bond in, for example, an oxime (Chapter 4), and we have now seen how an alkylidenecyclohexane such as **6** can be chiral without the need for a stereogenic carbon. Can one therefore obtain a chiral molecule in which the alkene part of **6** is replaced by, say, an oxime? Oxime **8** has four different groups in the key positions. The nature of the oxime group, in particular the lone pair on N, makes it impossible to construct a stable chiral oxime with only two different groups.

Make a model of the oxime **8** and also of its mirror image. Show that they are not superimposable, and note that indeed oximes such as **8**, that have no stereogenic centre, have been shown to exist as stable enantiomers.

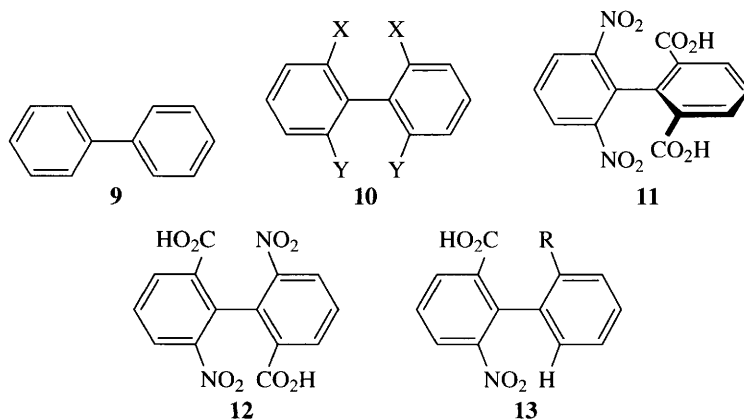
5.2 Biphenyls

Another class of molecules that does not possess stereogenic centres, and which when appropriately substituted can be chiral and resolvable, is the biphenyls; biphenyl itself is shown in **9**. We consider biphenyls with only *ortho* substituents and these can be chiral if two conditions are met. It is tacitly assumed that the substituents do not themselves contribute to any chirality, *i.e.* they do not contain stereogenic centres. These two conditions are:

1. Resolvable biphenyls must each contain bulky *ortho* substituents of sufficient size that rotation around the single bond that connects the aromatic rings is prevented by the 'collision' of substituents. If rotation around the single bond does occur, then one conformation, that in which the two aromatic rings are co-planar as in **10**, has a plane of symmetry and this rules out chirality.
2. Resolvable biphenyls must contain two *different* *ortho* substituents on *each* ring (hydrogen can, of course, count as a substituent, though it is diminutive). If one, or both, rings contain two identical substituents, the molecule is not chiral. In order to make this point clear, consider the molecule **11** and make a model of it. In **11** the molecule is represented in a key conformation with the two aromatic rings mutually perpendicular.

Ortho-substituted biphenyls also do not possess a stereogenic centre, but exist as stable enantiomers if (a) the substituents are large enough and (b) the *ortho* sites on each ring contain different substituents.

The problems outlined in condition (2) can be overcome by interchanging one NO₂ group in **11** for a CO₂H group; this gives the isomeric molecule **12**.



With the aid of the model of **11**, verify:

- (a) that there is a plane of symmetry that passes through the six carbons of the left-hand ring, and also through C(1) and C(4) of the right-hand ring;
- (b) that there is a corresponding plane, at right angles to the plane described in (a), that passes through six carbons of the right-hand ring and through C(1) and C(4) of the left-hand ring.

Appropriately substituted biphenyls are enantiomeric because of hindered rotation around a single bond, and are known as atropisomers.

Make a model of compound **12** and to your satisfaction demonstrate that the three-planes-of-symmetry obstacle to chirality, outlined in (1) and (2) and the exercises (a) and (b) above, has been overcome. It may be difficult to model a nitro group, but any sizeable alternative will do instead. Compound **12** was the first chiral substituted biphenyl to be resolved into enantiomers.

An enantiomerically pure biphenyl will racemize if the *ortho* substituents are insufficiently large.

The demonstration of chirality in **12**, and in other similarly substituted compounds, represents a new situation in that the chirality is caused by restricted rotation around a single bond [see condition (1) above], subject to the substituent patterns in the biphenyl [see condition (2) above]. The names given generally to isomers so derived are **atropisomers**, and to the phenomenon **atropisomerism**. ‘Atropisomer’ is derived from the Greek, ‘without turning’. Hindrance to rotation at normal working temperatures may be very severe, in which case the atropisomers will be configurationally stable indefinitely. Alternatively, racemization may occur over time if the sizes of the relevant *ortho* substituents are such that they can slip past each other and permit rotation around the carbon–carbon single bond that connects the aromatic rings. If this happens, condition (1) above is no longer fulfilled, because there is a plane of symmetry in the conformation in which both aromatic rings are co-planar. Rates of racemization have been measured for certain combinations of substituents, in the case of **13**, with variable R; half-life times for the first-order racemizations are given in Table 5.1. As might be expected, there is a clear indication from the data in Table 5.1 that racemization is easier the less bulky the substituent.

Table 5.1 Half-life(s) for racemization, $t_{1/2}(\text{rac})$, of substituted biphenyls **13**

<i>R</i>	<i>T</i> (°C)	$t_{1/2}(\text{rac})$ (min)
Me	118	179
NO ₂	118	125
CO ₂ H	118	91
OMe	25	9.4

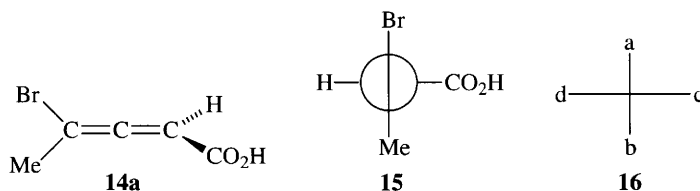
5.3 Absolute Configuration of Allenes and Biphenyls

5.3.1 Allenes

In order to describe the absolute configuration of chiral allenes, the Cahn–Ingold–Prelog (CIP) rules are first used to ascertain the priorities at the each terminal carbon, as is done for alkenes; at this point the similarity with alkenes ends. The method will be demonstrated with the trisubstituted allene **14a**:

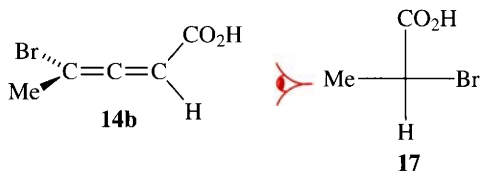
1. Look along the three-carbon allene unit from *one end* (either end can be chosen), and put uppermost the higher priority group at the ‘near’ end. Initially we look from the left in this example, which already has the ‘front’ substituents in a vertical plane.
2. Represent **14a** as a projection **15**, with the ‘near’ substituents vertical, and the ‘back’ substituents horizontal.
3. The ‘front’ substituents, prioritized, take precedence over the ‘back’ substituents, themselves prioritized. This gives $\text{Br} > \text{Me}$; $\text{CO}_2\text{H} > \text{H}$.
4. The overall sequence is as shown in **16**. Track this sequence alphabetically, noting especially the angular direction taken in going from ‘b’ to ‘c’, *i.e.* from the lower priority front substituent to the higher priority back substituent. If the sense of turn is clockwise, the allene has *R* configuration, whereas if the turn is anti-clockwise the allene has *S* configuration. The enantiomer shown in **14a** therefore has *S* configuration.

Allene configurations are determined with the aid of the CIP convention. Prioritize the substituents at each end separately. Look down the allene from either end, such that the near higher priority group, a, is at the top; the near lower priority group, b, is at the bottom. Groups c and d are horizontal at the back. Track the groups alphabetically; a clockwise turn from b to c gives *R* configuration.



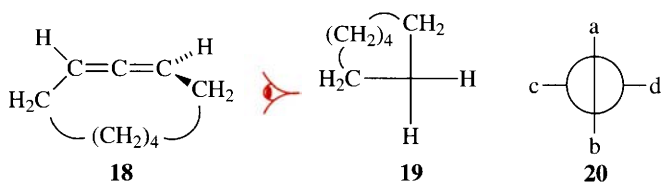
The last step is an adaptation of the original protocol (see Cahn³) and the method can also be conveniently applied to chiral biphenyls.

To assign configuration with the carboxylic acid group as a ‘near’ substituent, proceed as follows. Place the eye at the right-hand side of **14a** as drawn, and look toward and along the three-atom allene unit. Rotate the molecule clockwise around the axis defined by the three allenic carbons, in order to place the higher ranked ‘near’ substituent uppermost. The molecule now looks as in **14b**, with corresponding projection in **17**. Substituent ranking is as in **16**, and one again finds that the allene has *S* configuration.



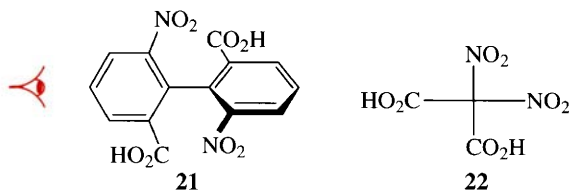
Configurations of biphenyls are determined the same way as for allenes; the *ortho* substituents must be included.

Just as the smallest carbocyclic ring that can incorporate a *trans* alkene is eight membered (Chapter 4), so the smallest sized ring that can contain an allene has nine carbons. One enantiomer of this compound is **18**. Rotate the molecule clockwise around the allene axis in order to place the higher ranked 'near' substituent uppermost. The projection from the right-hand side now has the form **19**, which is of general type **20**. Tracking the substituents alphabetically involves a clockwise turn between substituents 'b' and 'c'; hence the enantiomer of cyclonona-1,2-diene shown in **18** has *R* configuration.



5.3.2 Biphenyls

Consider the case of the first chiral biphenyl to be resolved, one enantiomer of which is shown in **21**. To determine the configuration of this enantiomer, one draws a projection formula similar to that previously shown for allenes, but with inclusion of relevant *ortho* groups; this is shown in **22**. If one tracks the substituents alphabetically, then the sense of turn between 'b' and 'c' is anti-clockwise, as in **16**, and so **22** has *S* configuration. It is possible to construct molecules that additionally contain substituents at the unsubstituted aromatic carbons of **21**; however, any additional substituents are not relevant to the question of atropisomerism.



5.4 Hexahelicene

Hexahelicene (**23**) exists in two enantiomeric forms because of overcrowding, which affects the end two rings. The congestion is relieved